

THE K-KNUTH SHAPE-INTERVAL CONJECTURE HOLDS ON ALPHABETS OF SIZE AT MOST EIGHT: AN EXACT FINITE CERTIFICATION

ZIJIAN ZENG

ABSTRACT. Gaetz et al. conjectured that the set of shapes occurring in a K-Knuth equivalence class of straight increasing tableaux is order-convex in Young’s lattice, and reported verification for alphabets of size at most seven. We give an exact finite certification for the first previously unreported endpoint, alphabet size eight. The computation enumerates all 9,069,306 increasing tableaux on [8], including 6,773,991 initial tableaux, constructs the exact K-Knuth congruence by the published finite algorithm, and checks all 988,384 initial classes. Every class is shape-interval complete. An independent verifier reconstructs the state space, transitions, components, and order-convexity test without importing discovery output; it also reproduces all published counts through seven letters. This is a certified finite result, not a proof for unbounded alphabets.

1. INTRODUCTION

K-Knuth equivalence is generated on words by the two classical Knuth moves together with idempotence and the K-theoretic braid move. Unlike ordinary Knuth equivalence, one K-Knuth class can contain row words of several increasing tableaux, and those tableaux can have different shapes. Gaetz et al. [1] asked whether the resulting shape set is nevertheless order-convex in Young’s lattice. Their Conjecture 7.6 states precisely this, and the sentence following the conjecture reports a computation for alphabets $[n]$ with $n \leq 7$.

The present result treats $n = 8$, the first endpoint not reported there. The calculation is entirely finite and integer-valued. Its trust boundary consists of a short serialized certificate, a strict parser, and an independent program that regenerates all mathematical objects. No floating-point optimization, random sampling, or unverified cached component data is used.

A search of the source paper, its latest arXiv version, indexed citing works, and exact-result phrases was performed through August 29, 2026. Within that recorded search scope, we found no earlier public $n = 8$ result. This is a bounded novelty statement, not a claim about every possible unpublished or unindexed source.

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2. DEFINITIONS AND THE FINITE ENDPOINT

A straight shape is a finite order ideal in $\mathbb{N}^2$, identified with a partition $\lambda = (\lambda_1 \geq \lambda_2 \geq \dots)$. We write $\lambda \subseteq \nu$ for coordinatewise inclusion and

$$[\lambda, \nu] = \{\mu : \lambda \subseteq \mu \subseteq \nu\}$$

for the closed interval in Young's lattice. An increasing tableau of shape λ is a filling that strictly increases along rows and down columns. It is initial on $[n] = \{1, \dots, n\}$ if every letter of $[n]$ occurs.

The row word $\text{row}(T)$ is read left-to-right within each row, from the bottom row upward. K-Knuth equivalence is the least monoid congruence containing

$$\begin{aligned} xzy &\equiv zxy, & yxz &\equiv yzx & (x < y < z), \\ x &\equiv xx, & xyx &\equiv yxy. \end{aligned}$$

Two tableaux are equivalent when their row words are. For a tableau class $\mathcal{C}$, set

$$\Sigma(\mathcal{C}) = \{\text{sh}(T) : T \in \mathcal{C}\}.$$

Order-preserving standardization commutes with all four relations and preserves shape. Consequently it is enough to work with initial tableaux on $[n]$. Moreover, the entry in box (i, j) is at least $i + j - 1$, so every shape lies in the finite staircase $\delta_n = (n, n - 1, \dots, 1)$.

3. EXACT ENUMERATION AND CONGRUENCE CLOSURE

3.1. State enumeration. The independent verifier first enumerates every partition $\lambda \subseteq \delta_n$. For a fixed shape it visits cells in row-major order. At (i, j) it tries every integer from

$$\max\{i + j - 1, T(i, j - 1) + 1, T(i - 1, j) + 1\} \quad \text{to} \quad n,$$

with nonexistent neighbors omitted. This is exhaustive and duplicate-free: the lower bound is necessary and sufficient at the current cell, and every tableau has a unique shape and filling.

Cells are serialized in an exact 80-bit code. On diagonal $d = i + j - 1$, digit zero denotes an absent cell and the remaining digits encode the possible entries $d, \dots, 8$. The code is decoded and tableau validity is checked before it is used.

3.2. K-Knuth components. For every state T and letter $a \in [n]$, the verifier performs Hecke insertion on an explicit integer grid and records $T \leftarrow a$. It then instantiates all primitive pairs

$$p \leftrightarrow pp, \quad pqp \leftrightarrow qpq \quad (p < q),$$

and the two Knuth pairs for every $x < y < z$. There are

$$n + \binom{n}{2} + 2\binom{n}{3}$$

such pairs. Their images at every state are merged, and every newly merged pair is closed under right insertion by every letter until the queue is empty. This is Algorithm 1 of Gaetz et al.; their Theorem 3.1 proves that its final components are exactly the tableau K-Knuth classes [1].

3.3. Shape intervals. For a finite poset and subset S , order-convexity is equivalent to

$$(\uparrow S) \cap (\downarrow S) \subseteq S.$$

The verifier assigns one bit to every shape in δ_n , constructs exact upward- and downward-closure bitsets, and tests

$$(\uparrow \Sigma(\mathcal{C})) \cap (\downarrow \Sigma(\mathcal{C})) \setminus \Sigma(\mathcal{C}) = \emptyset$$

for every initial component.

4. CERTIFIED RESULT

Theorem 4.1. *For every $0 \leq n \leq 8$ and every K-Knuth equivalence class $\mathcal{C}$ of initial straight increasing tableaux on $[n]$, if $\lambda, \nu \in \Sigma(\mathcal{C})$ and $\lambda \subseteq \nu$, then*

$$[\lambda, \nu] \subseteq \Sigma(\mathcal{C}).$$

Equivalently, Conjecture 7.6 holds for every ordered alphabet of cardinality at most eight.

Proof. The staircase bound makes the state space finite. The shape-first procedure above enumerates it exhaustively, and exact grid insertion gives every right transition. Algorithm 1 and Theorem 3.1 of [1] identify the computed components with K-Knuth classes. Each primitive move preserves its set of distinct letters, so filtering by the exact alphabet bitset selects whole initial components. Finally, the verifier finds the displayed closure difference empty for every such component. The strict certificate driver compares all totals in Table 1, as well as the $n = 8$ loop counts, and rejects any mismatch. Hence every computed shape set is order-convex. Standardization gives the assertion for every ordered alphabet of cardinality at most eight. $\square$

TABLE 1. Exact independently verified totals.

n	all tableaux	initial tableaux	initial classes	URTs	interval complete
0	1	1	1	1	yes
1	2	1	1	1	yes
2	6	3	3	3	yes
3	26	13	13	13	yes
4	162	87	79	71	yes
5	1,450	849	620	459	yes
6	18,626	11,915	6,036	3,313	yes
7	343,210	238,405	70,963	25,904	yes
8	9,069,306	6,773,991	988,384	215,295	yes

At $n = 8$, the verifier uses 148 primitive rules at each of 9,069,306 states, for 1,342,257,288 primitive tests. It records 7,303,139 successful component merges and 58,425,112 right-closure tests. The rows through $n = 7$ reproduce the initial-tableau, class, and URT counts in the source paper [1].

5. A LOCAL FORMULATION AND THE REMAINING GENERAL GAP

The finite data suggest the following prescribed-cover statement: if $T \equiv U$, $\lambda = \text{sh}(T) \subseteq \text{sh}(U) = \nu$, and b is any addable box of λ with $\lambda \cup \{b\} \subseteq \nu$, then the class should contain a tableau of shape $\lambda \cup \{b\}$. This local statement is equivalent to the general shape-interval conjecture by induction along a saturated chain in Young’s lattice. Theorem 4.1 verifies every instance with at most eight letters.

It is not yet a general proof. A K-jdt slide can change several boxes, and the source paper gives an example in which one K-Knuth move changes insertion shape by more than one box [1]. We do not know how to construct a tableau filling a prescribed cover while retaining K-Knuth equivalence. This is the precise open step beyond the certified endpoint.

6. REPRODUCIBILITY, AUDIT, AND LIMITATIONS

The certificate is `certificates/n8_certificate.json`. The command

```
sh scripts/verify_n8.sh
```

compiles the independent verifier and reconstructs all endpoints $0 \leq n \leq 8$. It prints SHA-256 hashes of the certificate, parser, verifier source, and verifier binary. The command

```
python3 tests/test_fail_closed.py
```

checks rejection of a missing required field, an extra trusted field, type confusion between Booleans and integers, a duplicate JSON key, and a modified exact count, together with one accepted control.

The discovery program uses letter-layer enumeration and row-bitset encodings. The verifier instead uses shape-first cell filling, an 80-bit cell encoding, explicit-grid insertion, generic rule generation, and bitset order closures. It imports no discovery states, transitions, components, or interval outputs. The public historical Python 2 implementation linked from OEIS A261601 was used only as provenance and was not imported by either certification path.

OpenAI Codex assisted with literature retrieval, formalization, implementation, testing, and drafting. The exact verifier and certificate expose the finite mathematical claim independently of generated prose. No Lean, Coq, Isabelle, or other proof assistant was used. The result is limited to alphabets of cardinality at most eight and makes no claim that the general conjecture is proved or that the manuscript has undergone peer review.

REFERENCES

1. Christian Gaetz, Michelle Mastrianni, Rebecca Patrias, Hailee Peck, Colleen Robichaux, David Schwein, and Ka Yu Tam, *K-knuth equivalence for increasing tableaux*, The Electronic Journal of Combinatorics **23** (2016), no. 1, P1.40.

INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNIVERSITY,
KUALA LUMPUR, 56000, MALAYSIA

Email address: `zijianzeng@foxmail.com`

Email address: `1002266693@ucsiuniversity.edu.my`